Take Home Final Exam - COVID-19

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```
library(readr)
library(ggplot2)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(fpp2)
```

```
## -- Attaching packages --------------------------------------------- fpp2 2.4 --

## v fma       2.4     v expsmooth 2.3

##
```

```
library(TTR)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```
library(psych)
```

```
##
## Attaching package: 'psych'

## The following object is masked from 'package:car':
##
##     logit

## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
library(corrplot)
```

```
## corrplot 0.92 loaded
```

```
library(RColorBrewer)
library(tinytex)
data <- read_csv("~/Desktop/Desktop - Edwin's MacBook Pro/Academic/Fall 2021/MSDS 401/Programming Assig
```

```
## Rows: 61900 Columns: 12

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (5): dateRep, countriesAndTerritories, geoId, countryterritoryCode, cont...
## dbl (7): day, month, year, cases, deaths, popData2019, Cumulative_number_for...

##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
covid_19 <- na.omit(data)
```

```
# 1) Descriptive Statistics
summary(covid_19)
```

```
##    dateRep               day             month             year
##  Length:58759       Min.   : 1.00   Min.   : 1.000   Min.   :2020
##  Class :character   1st Qu.: 8.00   1st Qu.: 5.000   1st Qu.:2020
##  Mode  :character   Median :15.00   Median : 7.000   Median :2020
##                     Mean   :15.65   Mean   : 7.262   Mean   :2020
##                     3rd Qu.:23.00   3rd Qu.:10.000   3rd Qu.:2020
##                     Max.   :31.00   Max.   :12.000   Max.   :2020
##      cases             deaths         countriesAndTerritories    geoId
##  Min.   : -8261   Min.   :-1918.00   Length:58759            Length:58759
##  1st Qu.:     0   1st Qu.:    0.00   Class :character        Class :character
##  Median :    20   Median :    0.00   Mode  :character        Mode  :character
##  Mean   :  1216   Mean   :   27.44
```

```
##  3rd Qu.:   312   3rd Qu.:    5.00
##  Max.   :234633   Max.   : 4928.00
##  countryterritoryCode  popData2019        continentExp
##  Length:58759         Min.   :8.150e+02   Length:58759
##  Class :character     1st Qu.:1.293e+06   Class :character
##  Mode  :character     Median :7.813e+06   Mode  :character
##                       Mean   :4.138e+07
##                       3rd Qu.:2.861e+07
##                       Max.   :1.434e+09
##  Cumulative_number_for_14_days_of_COVID-19_cases_per_100000
##  Min.   :-147.420
##  1st Qu.:   0.762
##  Median :   6.699
##  Mean   :  66.471
##  3rd Qu.:  52.618
##  Max.   :1900.836
```

```
summary(covid_19$cases)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -8261       0      20    1216     312  234633
```

```
quantile((covid_19$cases))
```

```
##     0%    25%    50%    75%   100%
##  -8261      0     20    312 234633
```

```
summary(covid_19$deaths)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -1918.00     0.00     0.00    27.44     5.00  4928.00
```

```
quantile((covid_19$deaths))
```

```
##    0%   25%   50%   75%  100%
## -1918     0     0     5  4928
```

```
summary(covid_19$`Cumulative_number_for_14_days_of_COVID-19_cases_per_100000`)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## -147.420    0.762    6.699   66.471   52.618 1900.836
```

```
quantile((covid_19$`Cumulative_number_for_14_days_of_COVID-19_cases_per_100000`))
```

```
##           0%          25%          50%          75%         100%
## -147.4195871    0.7620104    6.6988604   52.6182506 1900.8362104
```

The summary statistics for the COVID-19 dataset display summary values for the dataset's attributes. The numeric variables are divided into four quarters with the 1st quartile displaying the 25th percentile with 25% of the data points contained below Q1.The 2nd quartile displays the median values for the dataset's variables with 50% of the datapoints contained below Q2 and 50% of the datapoints contained above Q2.The 3rd quartile contains 75% of the datapoints below Q3 and 25% of the datapoints above Q3. The interquartile range contains the middle 50% of the datapoints distribution. The median value for the cases variable is 20 and the 3rd quartile value is 312. Thus for the cases variable, the middle value is 20 with values +-20 contained in the area 50% below 20 and 50% above 20. All values above 312 are contained in the 25% area above this value for the 3rd quartile. The median value for the deaths variable is 0 and the 3rd quartile value is 5. Thus for the deaths variable, the middle value is 0 with values +-0 contained in the area 50% below 20 and 50% above 20. All values above 5 are contained in the 25% area above this value for the 3rd quartile. Of note, the average value for the deaths variable is 27.44. The median value for the Cumulative_number_for_14_days_of_COVID-19_cases_per_100000 variable is 6.699 and the 3rd quartile value is 52.618. Thus for the Cumulative_number_for_14_days_of_COVID-19_cases_per_100000 variable, the middle value is 6.699 with values +-6.699 contained in the area 50% below 6.699 and 50% above 6.699. All values above 52.618 are contained in the 25% area above this value for the 3rd quartile.

```
range <- function(x) {max(x, na.rm = TRUE) - min(x, na.rm = TRUE)}
Global_incidence_rate <- range((covid_19$cases / covid_19$popData2019) * 100000)
Global_fatality_rate <- (range(covid_19$deaths) / range(covid_19$cases))
print(list('Global Incidence Rate:', Global_incidence_rate, 'Global Fatality Rate:', Global_fatality_ra
```

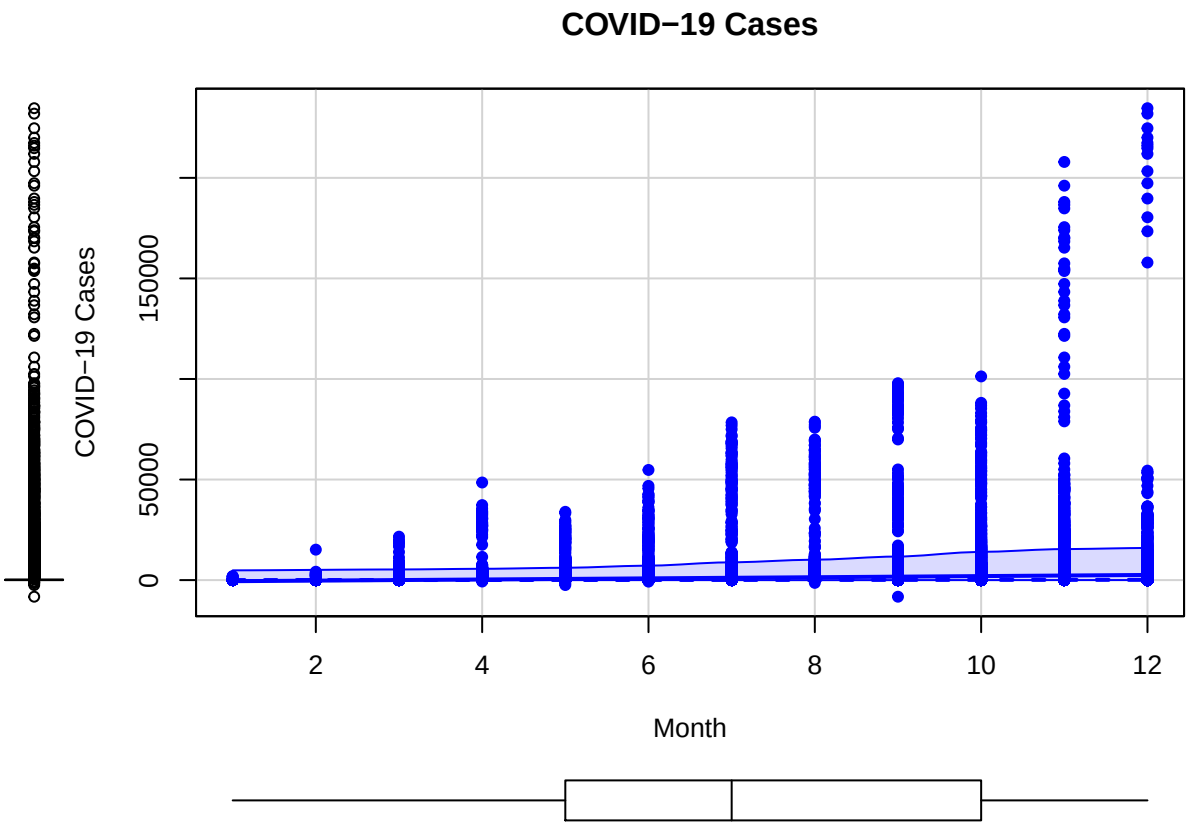
```
## [[1]]
## [1] "Global Incidence Rate:"
##
## [[2]]
## [1] 1084.505
##
## [[3]]
## [1] "Global Fatality Rate:"
##
## [[4]]
## [1] 0.02818513
```

The global incidence rate for COVID-19 is 1,084.05 which represents the daily total number of COVID-19 cases reported per 100,000 people. The global fatality rate for COVID-19 is 2.819% which represents the percentage of COVID-19 reported cases resulting in death.

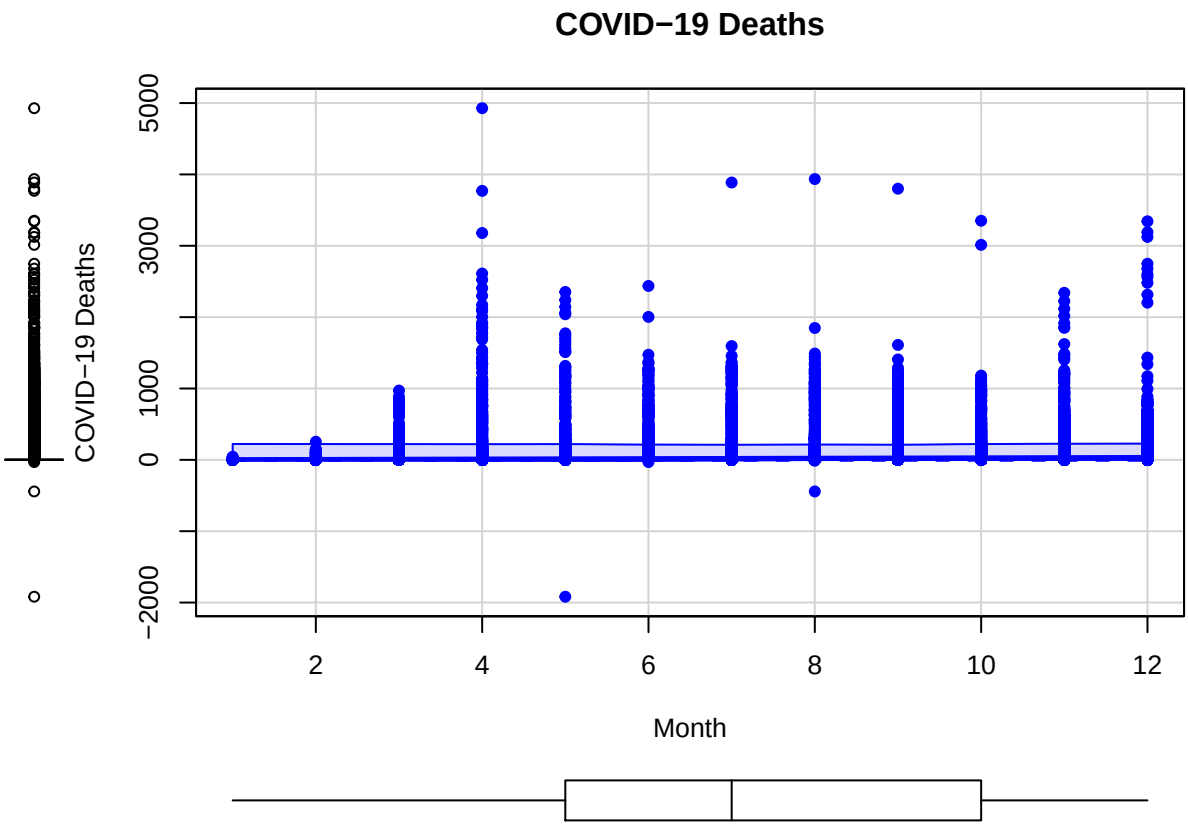
Including Visualizations

You can also embed plots, for example:

```
scatterplot(covid_19$month, covid_19$cases, main="COVID-19 Cases",
   xlab="Month", ylab="COVID-19 Cases", pch=19)
```




```
scatterplot(covid_19$month, covid_19$deaths, main="COVID-19 Deaths",
   xlab="Month", ylab="COVID-19 Deaths", pch=19)
```



The first scatter plot visualizes the aggregated count of cases for COVID-19 during 2020. Between the second and fourth months of 2020, the number of cases tripled. Between the fourth and sixth months of 2020, the number of cases slightly increased but the largest increases in cases occurred in the eleventh month by doubling from the amount of cases which occurred in month ten. The aggregated total count for cases during 2020 was 234,633 reflecting the maximum value for the cases variable. The second scatter plot visualizes the aggregated count of deaths for COVID-19 during 2020. The maximum fatality value occurred during the fourth month of 2020 with a total value of 4,928 deaths. The seventh, eight, and ninth months saw a consistent value of nearly 4,000 deaths per month.

```
# 2). Inferential Statistics
Canada <- covid_19 %>%
  filter(countriesAndTerritories == "Canada") %>%
  print()
```

```
## # A tibble: 337 x 12
##    dateRep      day month  year cases deaths countriesAndTerritories geoId
##    <chr>      <dbl> <dbl> <dbl> <dbl>  <dbl> <chr>                    <chr>
##  1 14/12/2020    14    12  2020  5891     81 Canada                   CA
##  2 13/12/2020    13    12  2020  6011     99 Canada                   CA
##  3 12/12/2020    12    12  2020  6772    142 Canada                   CA
##  4 11/12/2020    11    12  2020  6739    126 Canada                   CA
##  5 10/12/2020    10    12  2020  6295    116 Canada                   CA
##  6 09/12/2020     9    12  2020  5981     90 Canada                   CA
##  7 08/12/2020     8    12  2020  7872    112 Canada                   CA
##  8 07/12/2020     7    12  2020  6261     76 Canada                   CA
```

```
##  9 06/12/2020     6    12  2020  6352     93 Canada                  CA
## 10 05/12/2020     5    12  2020  6299     89 Canada                  CA
## # ... with 327 more rows, and 4 more variables: countryterritoryCode <chr>,
## #   popData2019 <dbl>, continentExp <chr>,
## #   Cumulative_number_for_14_days_of_COVID-19_cases_per_100000 <dbl>
```

```
USA <- covid_19 %>%
  filter(countriesAndTerritories == "United_States_of_America") %>%
  print()
```

```
## # A tibble: 337 x 12
##    dateRep      day month  year  cases deaths countriesAndTerritories  geoId
##    <chr>      <dbl> <dbl> <dbl>  <dbl>  <dbl> <chr>                     <chr>
##  1 14/12/2020    14    12  2020 189723   1340 United_States_of_America US
##  2 13/12/2020    13    12  2020 216017   2315 United_States_of_America US
##  3 12/12/2020    12    12  2020 234633   3343 United_States_of_America US
##  4 11/12/2020    11    12  2020 224680   2748 United_States_of_America US
##  5 10/12/2020    10    12  2020 220025   3124 United_States_of_America US
##  6 09/12/2020     9    12  2020 217344   2564 United_States_of_America US
##  7 08/12/2020     8    12  2020 197334   1433 United_States_of_America US
##  8 07/12/2020     7    12  2020 173432   1111 United_States_of_America US
##  9 06/12/2020     6    12  2020 211933   2203 United_States_of_America US
## 10 05/12/2020     5    12  2020 231930   2680 United_States_of_America US
## # ... with 327 more rows, and 4 more variables: countryterritoryCode <chr>,
## #   popData2019 <dbl>, continentExp <chr>,
## #   Cumulative_number_for_14_days_of_COVID-19_cases_per_100000 <dbl>
```

```
Canada_incidence_rate <- range((Canada$cases / Canada$popData2019) * 100000)
USA_incidence_rate <- range((USA$cases / USA$popData2019) * 100000)
Two_Countries_Incidence_test <- t.test((USA$cases / USA$popData2019) * 100000, (Canada$cases / Canada$po
Two_Countries_Incidence_test
```

```
##
##  Welch Two Sample t-test
##
## data:  (USA$cases/USA$popData2019) * 1e+05 and (Canada$cases/Canada$popData2019) * 1e+05
## t = 12.26, df = 393.7, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##   9.240322 12.769875
## sample estimates:
## mean of x mean of y
## 14.659608  3.654509
```

```
print(list('USA Incidence Rate:', USA_incidence_rate, 'Canada Incidence Rate:', Canada_incidence_rate))
```

```
## [[1]]
## [1] "USA Incidence Rate:"
##
## [[2]]
## [1] 71.30295
##
```

```
## [[3]]
## [1] "Canada Incidence Rate:"
##
## [[4]]
## [1] 21.04192
```

The null hypothesis is that no significant difference exist between the mean of the groups. The alternative hypothesis is that a significant difference exist between the mean of the groups. The average incidence rate for both groups is between 9.24 and 12.76 with a 95% confidence interval. As the p-value is smaller than 0.05, the results are statistically significant and the null hypothesis is rejected. The alternative hypothesis is accepted that there exist a significant difference between the mean of the groups for incidence rate.

```
Canada_fatality_rate <- (range(Canada$deaths) / range(Canada$cases))
USA_fatality_rate <- (range(USA$deaths) / range(USA$cases))
Two_Countries_Fatality_test <- t.test((Canada$deaths / Canada$cases), (USA$deaths / USA$cases), alterna
Two_Countries_Fatality_test
```

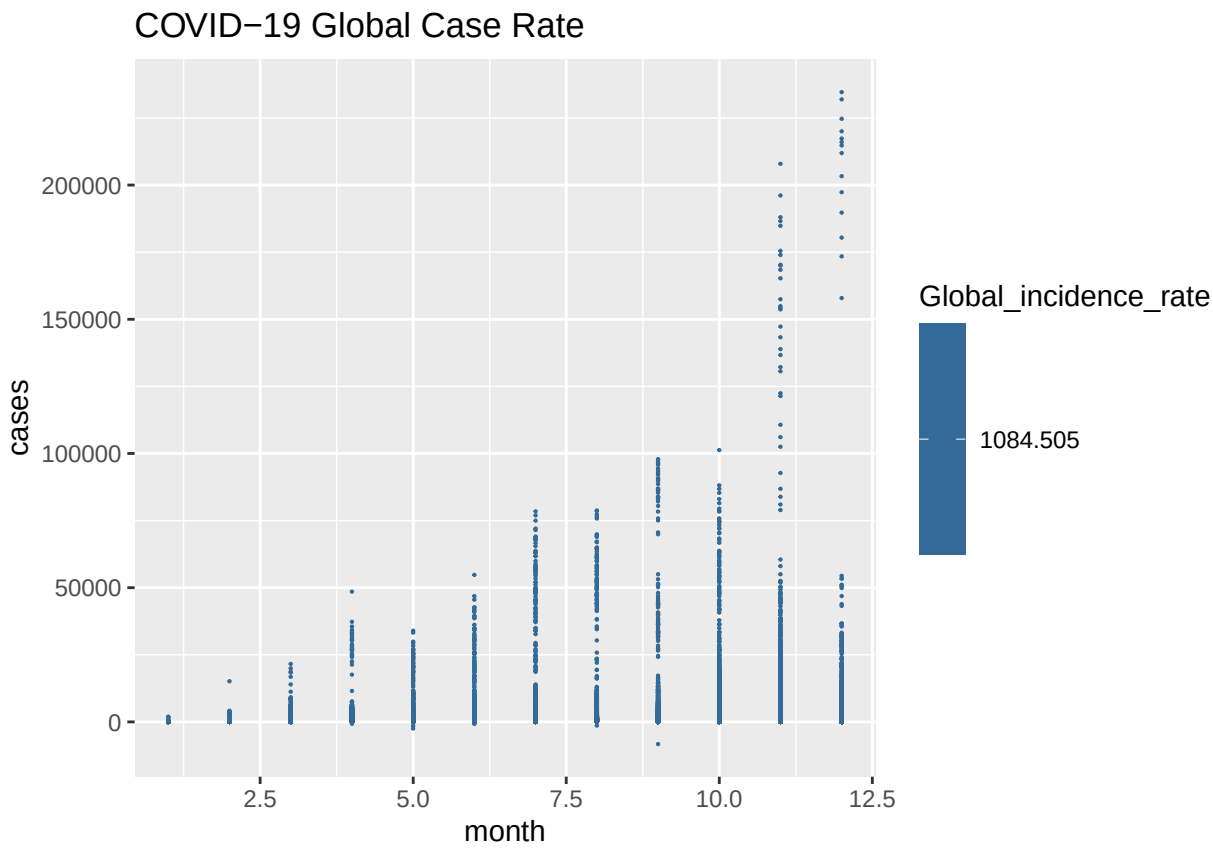
```
##
##  Welch Two Sample t-test
##
## data:  (Canada$deaths/Canada$cases) and (USA$deaths/USA$cases)
## t = 3.1477, df = 546.27, p-value = 0.001736
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.003839951 0.016588384
## sample estimates:
##  mean of x  mean of y
## 0.03784203 0.02762786
```

```
print(list('USA Fatality Rate:', USA_fatality_rate, 'Canada Fatality Rate:', Canada_fatality_rate))
```

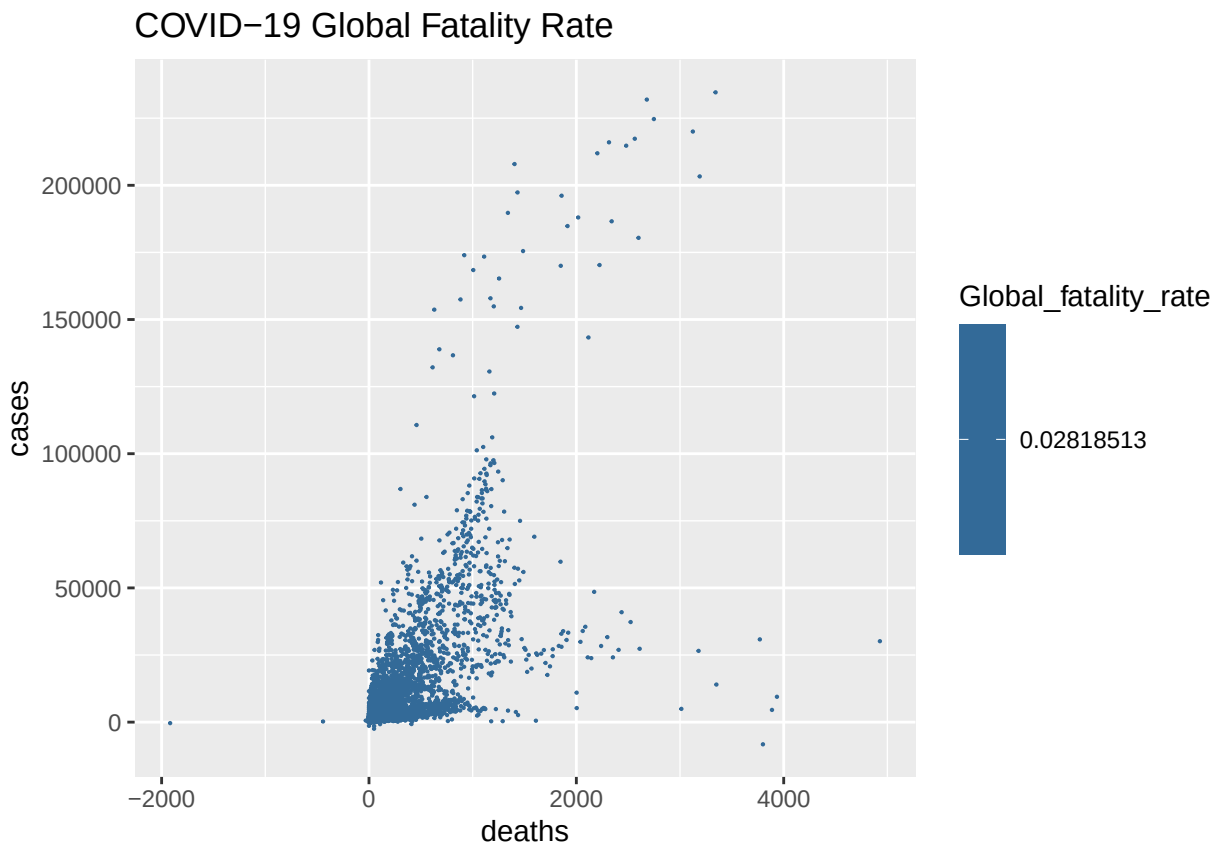
```
## [[1]]
## [1] "USA Fatality Rate:"
##
## [[2]]
## [1] 0.02100301
##
## [[3]]
## [1] "Canada Fatality Rate:"
##
## [[4]]
## [1] 0.02820122
```

The null hypothesis is that no significant difference exist between the mean of the groups. The alternative hypothesis is that a significant difference exist between the mean of the groups. The average incidence rate for both groups is between 0.0383 and 0.0165 with a 95% confidence interval. As the p-value is smaller than 0.05, the results are statistically significant and the null hypothesis is rejected. The alternative hypothesis is accepted that there exist a significant difference between the mean of the groups for fatality rates.

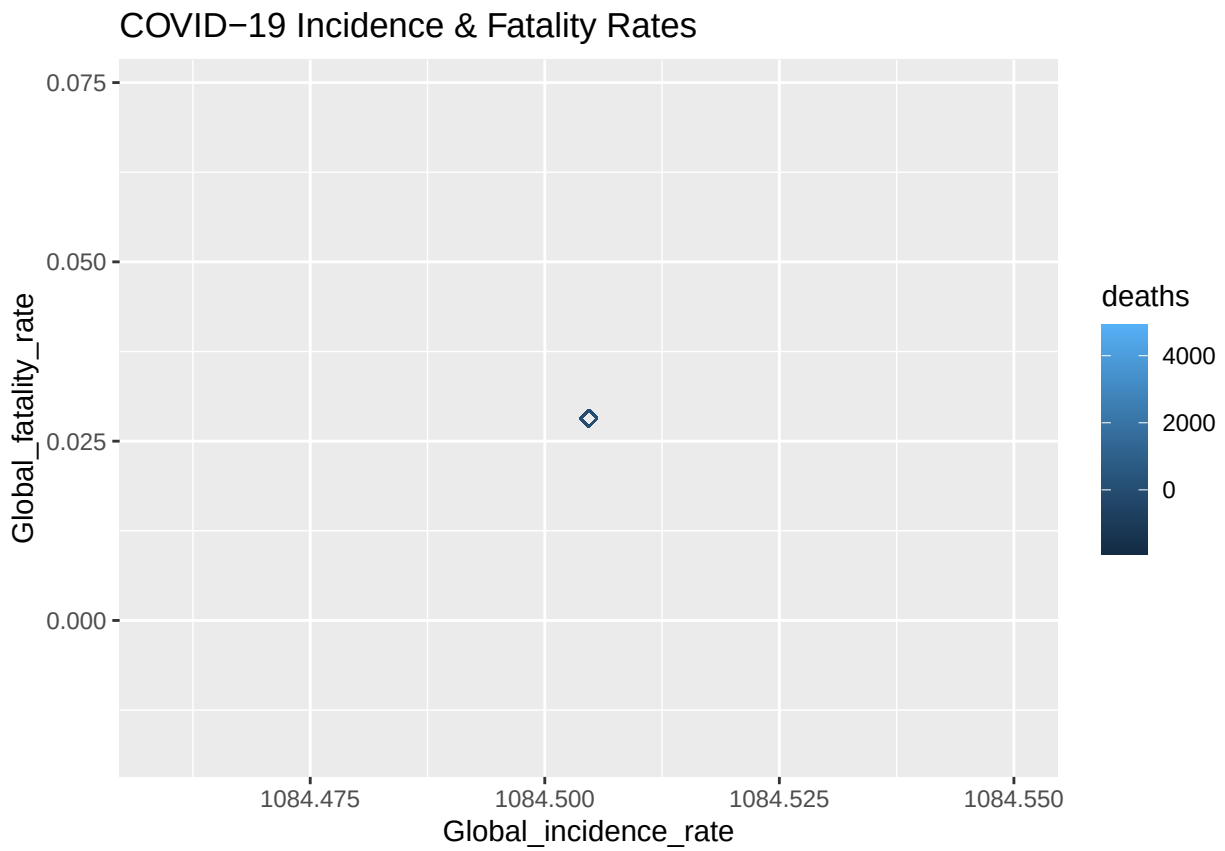
```
ggplot(data = covid_19, aes(x = month, y = cases)) + geom_point(aes(color = Global_incidence_rate), size
```




```
ggplot(data = covid_19, aes(x = deaths, y = cases)) + geom_point(aes(color = Global_fatality_rate), size
```




```
ggplot(data = covid_19, aes(x = Global_incidence_rate, y = Global_fatality_rate)) + geom_point(aes(colo
```




```
model <- lm(cases ~ popData2019 + deaths + countriesAndTerritories, data=covid_19)
summary(model)
```

```
##
## Call:
## lm(formula = cases ~ popData2019 + deaths + countriesAndTerritories,
##     data = covid_19)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -140304    -113      -3      14  144058
##
## Coefficients: (1 not defined because of singularities)
##                                                  Estimate Std. Error
## (Intercept)                                     3.240e+01  4.591e+02
## popData2019                                    -1.683e-06  1.525e-05
## deaths                                          3.025e+01  1.813e-01
## countriesAndTerritoriesAlbania                  4.026e+01  4.939e+02
## countriesAndTerritoriesAlgeria                  8.097e+01  3.752e+02
## countriesAndTerritoriesAndorra                 -1.417e+01  5.285e+02
## countriesAndTerritoriesAngola                   4.086e+01  3.270e+02
## countriesAndTerritoriesAnguilla                -3.234e+01  5.327e+02
## countriesAndTerritoriesAntigua_and_Barbuda     -3.215e+01  5.298e+02
## countriesAndTerritoriesArgentina                1.024e+03  4.088e+02
## countriesAndTerritoriesArmenia                  1.951e+02  4.800e+02
```

```
## countriesAndTerritoriesAruba                                -1.806e+01  5.305e+02
## countriesAndTerritoriesAustralia                             1.171e+01  2.952e+02
## countriesAndTerritoriesAustria                               5.372e+02  4.109e+02
## countriesAndTerritoriesAzerbaijan                            3.413e+02  3.998e+02
## countriesAndTerritoriesBahamas                              -2.120e+01  5.257e+02
## countriesAndTerritoriesBahrain                               2.043e+02  4.946e+02
## countriesAndTerritoriesBangladesh                            1.301e+03  2.088e+03
## countriesAndTerritoriesBarbados                             -3.173e+01  5.270e+02
## countriesAndTerritoriesBelarus                               3.546e+02  4.062e+02
## countriesAndTerritoriesBelgium                               1.798e+02  3.842e+02
## countriesAndTerritoriesBelize                               -1.833e+01  5.273e+02
## countriesAndTerritoriesBenin                                -5.787e+00  4.013e+02
## countriesAndTerritoriesBermuda                              -3.180e+01  5.302e+02
## countriesAndTerritoriesBhutan                               -2.946e+01  5.202e+02
## countriesAndTerritoriesBolivia                              -4.890e+02  4.027e+02
## countriesAndTerritoriesBonaire, Saint Eustatius and Saba -3.203e+01  5.342e+02
## countriesAndTerritoriesBosnia_and_Herzegovina               -2.645e+01  4.898e+02
## countriesAndTerritoriesBotswana                              1.801e+01  5.068e+02
## countriesAndTerritoriesBrazil                                4.521e+03  2.812e+03
## countriesAndTerritoriesBritish_Virgin_Islands               -3.217e+01  5.325e+02
## countriesAndTerritoriesBrunei_Darussalam                    -3.177e+01  5.234e+02
## countriesAndTerritoriesBulgaria                              6.812e+00  4.475e+02
## countriesAndTerritoriesBurkina_Faso                          8.912e+00  3.370e+02
## countriesAndTerritoriesBurundi                              -1.015e+01  4.091e+02
## countriesAndTerritoriesCambodia                             -3.548e+00  3.417e+02
## countriesAndTerritoriesCameroon                              5.549e+01  3.186e+02
## countriesAndTerritoriesCanada                                1.922e+02  3.276e+02
## countriesAndTerritoriesCape_Verde                           -9.287e-03  5.246e+02
## countriesAndTerritoriesCayman_Islands                       -3.130e+01  5.302e+02
## countriesAndTerritoriesCentral_African_Republic             -1.282e+01  4.741e+02
## countriesAndTerritoriesChad                                 -1.070e+01  3.675e+02
## countriesAndTerritoriesChile                                 3.349e+02  3.418e+02
## countriesAndTerritoriesChina                                 2.229e+03  2.145e+04
## countriesAndTerritoriesColombia                              9.708e+02  4.683e+02
## countriesAndTerritoriesComoros                              -2.893e+01  5.342e+02
## countriesAndTerritoriesCongo                                -1.165e+01  4.670e+02
## countriesAndTerritoriesCosta_Rica                            3.239e+02  4.685e+02
## countriesAndTerritoriesCote_dIvoire                          7.773e+01  3.192e+02
## countriesAndTerritoriesCroatia                               2.620e+02  4.655e+02
## countriesAndTerritoriesCuba                                  6.807e+00  4.050e+02
## countriesAndTerritoriesCuraçao                              -1.904e+01  5.301e+02
## countriesAndTerritoriesCyprus                                1.672e+01  5.183e+02
## countriesAndTerritoriesCzechia                               8.473e+02  3.922e+02
## countriesAndTerritoriesDemocratic_Republic_of_the_Congo      1.282e+02  9.546e+02
## countriesAndTerritoriesDenmark                               2.186e+02  4.448e+02
## countriesAndTerritoriesDjibouti                             -1.581e+01  5.189e+02
## countriesAndTerritoriesDominica                             -3.199e+01  5.309e+02
## countriesAndTerritoriesDominican_Republic                    2.396e+02  3.932e+02
## countriesAndTerritoriesEcuador                              -6.585e+02  3.341e+02
## countriesAndTerritoriesEgypt                                -1.269e+02  1.146e+03
## countriesAndTerritoriesEl_Salvador                          -6.082e-01  4.560e+02
## countriesAndTerritoriesEquatorial_Guinea                    -2.018e+01  5.133e+02
## countriesAndTerritoriesEritrea                              -2.381e+01  4.901e+02
## countriesAndTerritoriesEstonia                               1.064e+01  4.992e+02
```

```
## countriesAndTerritoriesEswatini                          -1.932e+01  5.158e+02
## countriesAndTerritoriesEthiopia                           3.925e+02  1.325e+03
## countriesAndTerritoriesFalkland_Islands_(Malvinas)       -3.236e+01  5.351e+02
## countriesAndTerritoriesFaroe_Islands                     -3.043e+01  5.269e+02
## countriesAndTerritoriesFiji                              -3.097e+01  5.202e+02
## countriesAndTerritoriesFinland                            2.827e+01  4.489e+02
## countriesAndTerritoriesFrance                             1.935e+03  6.670e+02
## countriesAndTerritoriesFrench_Polynesia                   1.780e+01  5.273e+02
## countriesAndTerritoriesGabon                             -4.167e-01  5.033e+02
## countriesAndTerritoriesGambia                            -2.810e+01  5.023e+02
## countriesAndTerritoriesGeorgia                            3.798e+02  4.665e+02
## countriesAndTerritoriesGermany                            2.102e+03  8.922e+02
## countriesAndTerritoriesGhana                              1.829e+02  3.202e+02
## countriesAndTerritoriesGibraltar                         -2.901e+01  5.306e+02
## countriesAndTerritoriesGreece                             3.008e+01  3.919e+02
## countriesAndTerritoriesGreenland                         -3.227e+01  5.303e+02
## countriesAndTerritoriesGrenada                           -3.198e+01  5.304e+02
## countriesAndTerritoriesGuam                              -1.852e+01  5.287e+02
## countriesAndTerritoriesGuatemala                         -1.999e+01  3.540e+02
## countriesAndTerritoriesGuernsey                          -3.293e+01  5.302e+02
## countriesAndTerritoriesGuinea                             3.118e+01  3.918e+02
## countriesAndTerritoriesGuinea_Bissau                     -2.484e+01  5.099e+02
## countriesAndTerritoriesGuyana                            -2.628e+01  5.202e+02
## countriesAndTerritoriesHaiti                             -3.818e+00  4.073e+02
## countriesAndTerritoriesHoly_See                          -3.231e+01  5.297e+02
## countriesAndTerritoriesHonduras                           7.647e+01  4.201e+02
## countriesAndTerritoriesHungary                            2.356e+02  4.174e+02
## countriesAndTerritoriesIceland                           -1.782e+01  5.104e+02
## countriesAndTerritoriesIndia                              1.878e+04  2.042e+04
## countriesAndTerritoriesIndonesia                          5.703e+02  3.716e+03
## countriesAndTerritoriesIran                              -1.289e+03  8.910e+02
## countriesAndTerritoriesIraq                               6.133e+02  3.420e+02
## countriesAndTerritoriesIreland                            1.149e+01  4.556e+02
## countriesAndTerritoriesIsle_of_Man                       -3.402e+01  5.302e+02
## countriesAndTerritoriesIsrael                             7.818e+02  4.151e+02
## countriesAndTerritoriesItaly                             -2.511e+02  5.778e+02
## countriesAndTerritoriesJamaica                           -1.432e+01  4.939e+02
## countriesAndTerritoriesJapan                              4.822e+02  1.540e+03
## countriesAndTerritoriesJersey                            -2.914e+01  5.297e+02
## countriesAndTerritoriesJordan                             5.820e+02  4.163e+02
## countriesAndTerritoriesKazakhstan                         4.147e+02  3.478e+02
## countriesAndTerritoriesKenya                              2.229e+02  4.940e+02
## countriesAndTerritoriesKosovo                             9.800e+00  5.078e+02
## countriesAndTerritoriesKuwait                             3.300e+02  4.640e+02
## countriesAndTerritoriesKyrgyzstan                         1.253e+02  4.564e+02
## countriesAndTerritoriesLaos                              -2.021e+01  4.502e+02
## countriesAndTerritoriesLatvia                             2.657e+01  5.045e+02
## countriesAndTerritoriesLebanon                            3.101e+02  4.336e+02
## countriesAndTerritoriesLesotho                           -2.426e+01  5.247e+02
## countriesAndTerritoriesLiberia                           -2.731e+01  4.722e+02
## countriesAndTerritoriesLibya                              1.834e+02  4.544e+02
## countriesAndTerritoriesLiechtenstein                     -2.886e+01  5.282e+02
## countriesAndTerritoriesLithuania                          1.859e+02  4.819e+02
## countriesAndTerritoriesLuxembourg                         5.741e+01  5.083e+02
```

```
## countriesAndTerritoriesMadagascar                              5.121e+01  3.209e+02
## countriesAndTerritoriesMalawi                                  8.318e-01  3.551e+02
## countriesAndTerritoriesMalaysia                                2.325e+02  2.998e+02
## countriesAndTerritoriesMaldives                                1.308e+01  5.220e+02
## countriesAndTerritoriesMali                                    8.954e-01  3.457e+02
## countriesAndTerritoriesMalta                                  -9.240e+00  5.217e+02
## countriesAndTerritoriesMarshall_Islands                       -3.221e+01  8.648e+02
## countriesAndTerritoriesMauritania                             -9.274e+00  4.763e+02
## countriesAndTerritoriesMauritius                              -2.963e+01  5.157e+02
## countriesAndTerritoriesMexico                                 -6.337e+03  1.551e+03
## countriesAndTerritoriesMoldova                                 1.567e+02  4.805e+02
## countriesAndTerritoriesMonaco                                 -3.047e+01  5.164e+02
## countriesAndTerritoriesMongolia                               -2.351e+01  4.915e+02
## countriesAndTerritoriesMontenegro                              6.041e+01  5.229e+02
## countriesAndTerritoriesMontserrat                             -3.247e+01  5.314e+02
## countriesAndTerritoriesMorocco                                 7.752e+02  3.428e+02
## countriesAndTerritoriesMozambique                              6.852e+01  3.239e+02
## countriesAndTerritoriesMyanmar                                 2.151e+02  5.138e+02
## countriesAndTerritoriesNepal                                   6.240e+02  2.966e+02
## countriesAndTerritoriesNetherlands                             9.143e+02  3.335e+02
## countriesAndTerritoriesNew_Caledonia                          -3.184e+01  5.278e+02
## countriesAndTerritoriesNew_Zealand                            -2.134e+01  4.583e+02
## countriesAndTerritoriesNicaragua                              -1.745e+01  4.550e+02
## countriesAndTerritoriesNiger                                   6.514e+00  3.278e+02
## countriesAndTerritoriesNigeria                                 4.193e+02  2.658e+03
## countriesAndTerritoriesNorth_Macedonia                         4.931e-03  4.906e+02
## countriesAndTerritoriesNorthern_Mariana_Islands               -3.188e+01  5.333e+02
## countriesAndTerritoriesNorway                                  6.060e+01  4.503e+02
## countriesAndTerritoriesOman                                    2.229e+02  4.555e+02
## countriesAndTerritoriesPakistan                                8.551e+02  2.895e+03
## countriesAndTerritoriesPalestine                               3.192e+02  4.695e+02
## countriesAndTerritoriesPanama                                  3.168e+02  4.781e+02
## countriesAndTerritoriesPapua_New_Guinea                       -1.574e+01  4.321e+02
## countriesAndTerritoriesParaguay                                1.097e+02  4.476e+02
## countriesAndTerritoriesPeru                                   -4.410e+02  3.243e+02
## countriesAndTerritoriesPhilippines                             7.059e+02  1.260e+03
## countriesAndTerritoriesPoland                                  1.670e+03  3.511e+02
## countriesAndTerritoriesPortugal                                6.434e+02  4.115e+02
## countriesAndTerritoriesPuerto_Rico                             2.488e+02  4.983e+02
## countriesAndTerritoriesQatar                                   3.739e+02  4.806e+02
## countriesAndTerritoriesRomania                                 4.524e+02  3.198e+02
## countriesAndTerritoriesRussia                                  3.875e+03  1.825e+03
## countriesAndTerritoriesRwanda                                  7.620e+00  3.931e+02
## countriesAndTerritoriesSaint_Kitts_and_Nevis                  -3.224e+01  5.319e+02
## countriesAndTerritoriesSaint_Lucia                            -3.151e+01  5.275e+02
## countriesAndTerritoriesSaint_Vincent_and_the_Grenadines       -3.185e+01  5.307e+02
## countriesAndTerritoriesSan_Marino                             -3.113e+01  5.146e+02
## countriesAndTerritoriesSao_Tome_and_Principe                  -2.996e+01  5.340e+02
## countriesAndTerritoriesSaudi_Arabia                            6.777e+02  3.297e+02
## countriesAndTerritoriesSenegal                                 1.924e+01  3.612e+02
## countriesAndTerritoriesSerbia                                  7.127e+02  4.479e+02
## countriesAndTerritoriesSeychelles                             -3.154e+01  5.285e+02
## countriesAndTerritoriesSierra_Leone                           -1.858e+01  4.457e+02
## countriesAndTerritoriesSingapore                               1.478e+02  4.448e+02
```

```
## countriesAndTerritoriesSint_Maarten                       -3.017e+01  5.321e+02
## countriesAndTerritoriesSlovakia                            3.386e+02  4.639e+02
## countriesAndTerritoriesSlovenia                            1.628e+02  5.024e+02
## countriesAndTerritoriesSolomon_Islands                    -3.108e+01  7.690e+02
## countriesAndTerritoriesSomalia                            -2.877e+00  3.702e+02
## countriesAndTerritoriesSouth_Africa                        6.489e+02  5.658e+02
## countriesAndTerritoriesSouth_Korea                         1.302e+02  4.622e+02
## countriesAndTerritoriesSouth_Sudan                        -8.246e+00  4.153e+02
## countriesAndTerritoriesSpain                               9.097e+02  4.146e+02
## countriesAndTerritoriesSri_Lanka                           8.945e+01  3.111e+02
## countriesAndTerritoriesSudan                              -3.383e+01  3.919e+02
## countriesAndTerritoriesSuriname                           -2.440e+01  5.239e+02
## countriesAndTerritoriesSweden                              2.602e+02  3.965e+02
## countriesAndTerritoriesSwitzerland                         6.041e+02  4.143e+02
## countriesAndTerritoriesSyria                              -2.909e+01  3.606e+02
## countriesAndTerritoriesTaiwan                              9.192e+00  2.989e+02
## countriesAndTerritoriesTajikistan                          2.985e+01  4.423e+02
## countriesAndTerritoriesThailand                            9.215e+01  7.032e+02
## countriesAndTerritoriesTimor_Leste                        -3.010e+01  5.159e+02
## countriesAndTerritoriesTogo                               -1.411e+01  4.371e+02
## countriesAndTerritoriesTrinidad_and_Tobago                -1.818e+01  5.123e+02
## countriesAndTerritoriesTunisia                            -3.678e+01  3.997e+02
## countriesAndTerritoriesTurkey                              2.025e+03  8.862e+02
## countriesAndTerritoriesTurks_and_Caicos_islands           -2.990e+01  5.318e+02
## countriesAndTerritoriesUganda                              1.237e+02  4.080e+02
## countriesAndTerritoriesUkraine                             1.713e+03  4.028e+02
## countriesAndTerritoriesUnited_Arab_Emirates                4.864e+02  4.025e+02
## countriesAndTerritoriesUnited_Kingdom                     -1.922e+02  6.621e+02
## countriesAndTerritoriesUnited_Republic_of_Tanzania         6.483e+01  5.608e+02
## countriesAndTerritoriesUnited_States_of_America            2.191e+04  4.608e+03
## countriesAndTerritoriesUnited_States_Virgin_Islands       -2.786e+01  5.310e+02
## countriesAndTerritoriesUruguay                            -1.045e+00  4.885e+02
## countriesAndTerritoriesUzbekistan                          2.395e+02  3.283e+02
## countriesAndTerritoriesVanuatu                            -3.189e+01  1.038e+03
## countriesAndTerritoriesVenezuela                           3.160e+02  3.187e+02
## countriesAndTerritoriesVietnam                             1.310e+02  1.088e+03
## countriesAndTerritoriesWestern_Sahara                     -2.810e+01  5.352e+02
## countriesAndTerritoriesYemen                              -5.216e+01  3.306e+02
## countriesAndTerritoriesZambia                              2.534e+01  3.537e+02
## countriesAndTerritoriesZimbabwe                                   NA         NA
##                                                           t value Pr(>|t|)
## (Intercept)                                                 0.071  0.94375
## popData2019                                                -0.110  0.91213
## deaths                                                    166.840  < 2e-16 ***
## countriesAndTerritoriesAlbania                              0.082  0.93503
## countriesAndTerritoriesAlgeria                              0.216  0.82916
## countriesAndTerritoriesAndorra                             -0.027  0.97862
## countriesAndTerritoriesAngola                               0.125  0.90056
## countriesAndTerritoriesAnguilla                            -0.061  0.95158
## countriesAndTerritoriesAntigua_and_Barbuda                 -0.061  0.95161
## countriesAndTerritoriesArgentina                            2.506  0.01221 *
## countriesAndTerritoriesArmenia                              0.406  0.68445
## countriesAndTerritoriesAruba                               -0.034  0.97284
## countriesAndTerritoriesAustralia                            0.040  0.96837
```

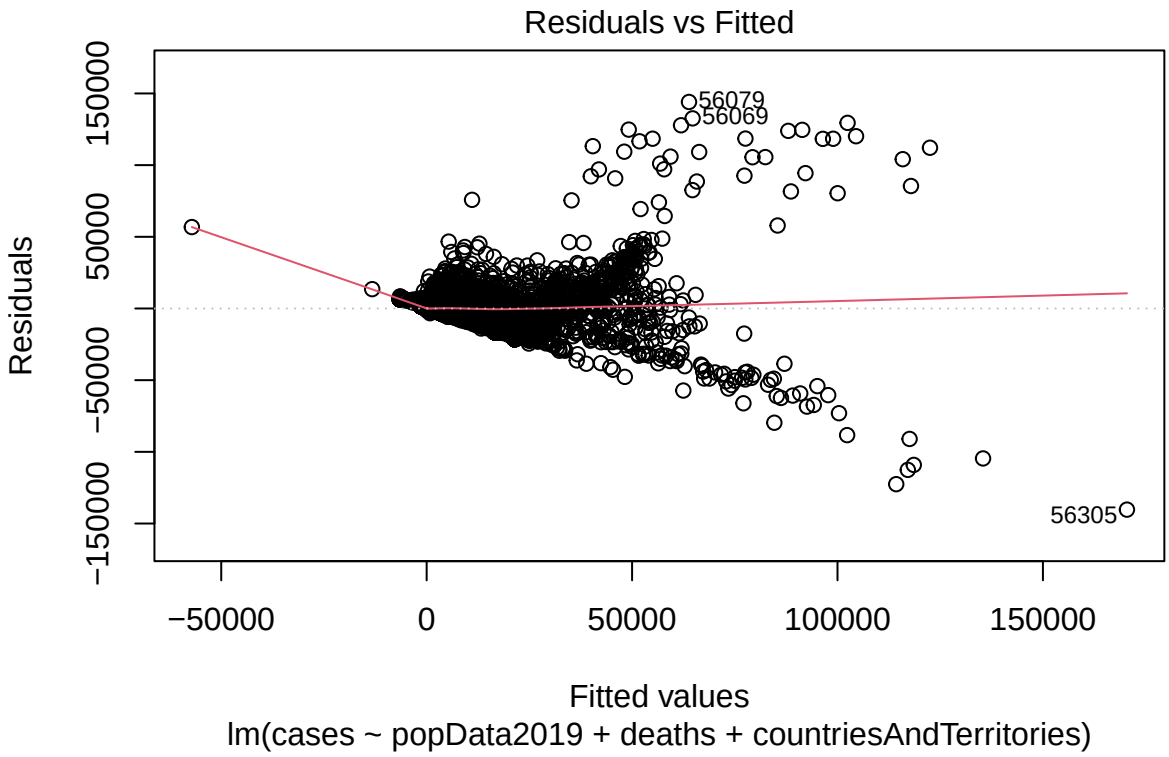
```
## countriesAndTerritoriesAustria                                1.307  0.19106
## countriesAndTerritoriesAzerbaijan                             0.854  0.39331
## countriesAndTerritoriesBahamas                               -0.040  0.96783
## countriesAndTerritoriesBahrain                                0.413  0.67950
## countriesAndTerritoriesBangladesh                             0.623  0.53335
## countriesAndTerritoriesBarbados                              -0.060  0.95199
## countriesAndTerritoriesBelarus                                0.873  0.38269
## countriesAndTerritoriesBelgium                                0.468  0.63986
## countriesAndTerritoriesBelize                                -0.035  0.97227
## countriesAndTerritoriesBenin                                 -0.014  0.98849
## countriesAndTerritoriesBermuda                               -0.060  0.95218
## countriesAndTerritoriesBhutan                                -0.057  0.95484
## countriesAndTerritoriesBolivia                               -1.214  0.22461
## countriesAndTerritoriesBonaire, Saint Eustatius and Saba     -0.060  0.95219
## countriesAndTerritoriesBosnia_and_Herzegovina                -0.054  0.95694
## countriesAndTerritoriesBotswana                               0.036  0.97166
## countriesAndTerritoriesBrazil                                 1.608  0.10795
## countriesAndTerritoriesBritish_Virgin_Islands                -0.060  0.95182
## countriesAndTerritoriesBrunei_Darussalam                     -0.061  0.95161
## countriesAndTerritoriesBulgaria                               0.015  0.98786
## countriesAndTerritoriesBurkina_Faso                           0.026  0.97890
## countriesAndTerritoriesBurundi                               -0.025  0.98020
## countriesAndTerritoriesCambodia                              -0.010  0.99171
## countriesAndTerritoriesCameroon                               0.174  0.86172
## countriesAndTerritoriesCanada                                 0.587  0.55735
## countriesAndTerritoriesCape_Verde                             0.000  0.99999
## countriesAndTerritoriesCayman_Islands                        -0.059  0.95293
## countriesAndTerritoriesCentral_African_Republic              -0.027  0.97843
## countriesAndTerritoriesChad                                  -0.029  0.97678
## countriesAndTerritoriesChile                                  0.980  0.32708
## countriesAndTerritoriesChina                                  0.104  0.91724
## countriesAndTerritoriesColombia                               2.073  0.03818 *
## countriesAndTerritoriesComoros                               -0.054  0.95682
## countriesAndTerritoriesCongo                                 -0.025  0.98009
## countriesAndTerritoriesCosta_Rica                             0.691  0.48927
## countriesAndTerritoriesCote_dIvoire                           0.244  0.80761
## countriesAndTerritoriesCroatia                                0.563  0.57361
## countriesAndTerritoriesCuba                                   0.017  0.98659
## countriesAndTerritoriesCuraçao                               -0.036  0.97135
## countriesAndTerritoriesCyprus                                 0.032  0.97426
## countriesAndTerritoriesCzechia                                2.160  0.03076 *
## countriesAndTerritoriesDemocratic_Republic_of_the_Congo       0.134  0.89319
## countriesAndTerritoriesDenmark                                0.491  0.62311
## countriesAndTerritoriesDjibouti                              -0.030  0.97570
## countriesAndTerritoriesDominica                              -0.060  0.95196
## countriesAndTerritoriesDominican_Republic                     0.609  0.54234
## countriesAndTerritoriesEcuador                               -1.971  0.04874 *
## countriesAndTerritoriesEgypt                                 -0.111  0.91182
## countriesAndTerritoriesEl_Salvador                           -0.001  0.99894
## countriesAndTerritoriesEquatorial_Guinea                     -0.039  0.96864
## countriesAndTerritoriesEritrea                               -0.049  0.96125
## countriesAndTerritoriesEstonia                                0.021  0.98299
## countriesAndTerritoriesEswatini                              -0.037  0.97013
## countriesAndTerritoriesEthiopia                               0.296  0.76701
```

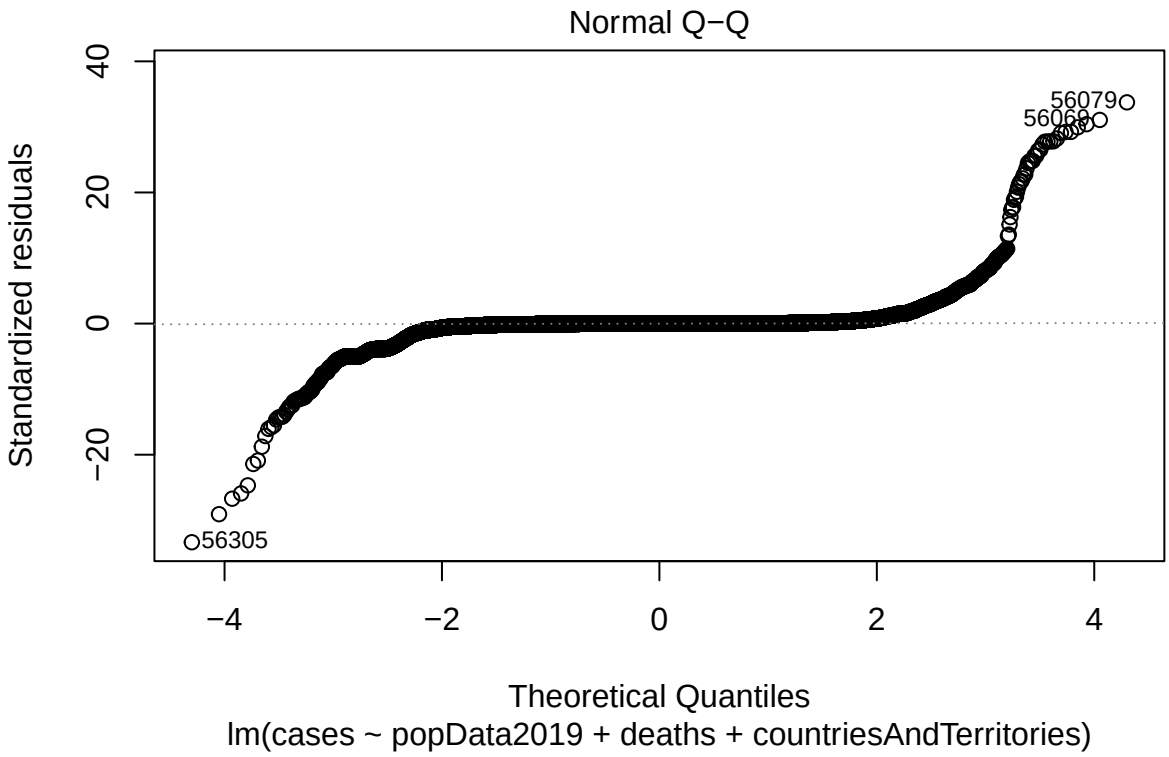
```
## countriesAndTerritoriesFalkland_Islands_(Malvinas)        -0.060  0.95178
## countriesAndTerritoriesFaroe_Islands                      -0.058  0.95394
## countriesAndTerritoriesFiji                               -0.060  0.95252
## countriesAndTerritoriesFinland                             0.063  0.94979
## countriesAndTerritoriesFrance                              2.902  0.00371 **
## countriesAndTerritoriesFrench_Polynesia                    0.034  0.97307
## countriesAndTerritoriesGabon                              -0.001  0.99934
## countriesAndTerritoriesGambia                             -0.056  0.95539
## countriesAndTerritoriesGeorgia                             0.814  0.41551
## countriesAndTerritoriesGermany                             2.357  0.01845 *
## countriesAndTerritoriesGhana                               0.571  0.56797
## countriesAndTerritoriesGibraltar                          -0.055  0.95639
## countriesAndTerritoriesGreece                              0.077  0.93881
## countriesAndTerritoriesGreenland                          -0.061  0.95148
## countriesAndTerritoriesGrenada                            -0.060  0.95192
## countriesAndTerritoriesGuam                               -0.035  0.97205
## countriesAndTerritoriesGuatemala                          -0.056  0.95498
## countriesAndTerritoriesGuernsey                           -0.062  0.95048
## countriesAndTerritoriesGuinea                              0.080  0.93656
## countriesAndTerritoriesGuinea_Bissau                      -0.049  0.96114
## countriesAndTerritoriesGuyana                             -0.051  0.95971
## countriesAndTerritoriesHaiti                              -0.009  0.99252
## countriesAndTerritoriesHoly_See                           -0.061  0.95136
## countriesAndTerritoriesHonduras                            0.182  0.85556
## countriesAndTerritoriesHungary                             0.565  0.57240
## countriesAndTerritoriesIceland                            -0.035  0.97215
## countriesAndTerritoriesIndia                               0.920  0.35780
## countriesAndTerritoriesIndonesia                           0.153  0.87802
## countriesAndTerritoriesIran                               -1.447  0.14791
## countriesAndTerritoriesIraq                                1.793  0.07297 .
## countriesAndTerritoriesIreland                             0.025  0.97988
## countriesAndTerritoriesIsle_of_Man                        -0.064  0.94885
## countriesAndTerritoriesIsrael                              1.883  0.05968 .
## countriesAndTerritoriesItaly                              -0.435  0.66391
## countriesAndTerritoriesJamaica                            -0.029  0.97687
## countriesAndTerritoriesJapan                               0.313  0.75412
## countriesAndTerritoriesJersey                             -0.055  0.95613
## countriesAndTerritoriesJordan                              1.398  0.16209
## countriesAndTerritoriesKazakhstan                          1.193  0.23305
## countriesAndTerritoriesKenya                               0.451  0.65189
## countriesAndTerritoriesKosovo                              0.019  0.98460
## countriesAndTerritoriesKuwait                              0.711  0.47701
## countriesAndTerritoriesKyrgyzstan                          0.275  0.78366
## countriesAndTerritoriesLaos                               -0.045  0.96420
## countriesAndTerritoriesLatvia                              0.053  0.95801
## countriesAndTerritoriesLebanon                             0.715  0.47449
## countriesAndTerritoriesLesotho                            -0.046  0.96313
## countriesAndTerritoriesLiberia                            -0.058  0.95389
## countriesAndTerritoriesLibya                               0.404  0.68654
## countriesAndTerritoriesLiechtenstein                      -0.055  0.95643
## countriesAndTerritoriesLithuania                           0.386  0.69965
## countriesAndTerritoriesLuxembourg                          0.113  0.91008
## countriesAndTerritoriesMadagascar                          0.160  0.87321
## countriesAndTerritoriesMalawi                              0.002  0.99813
```

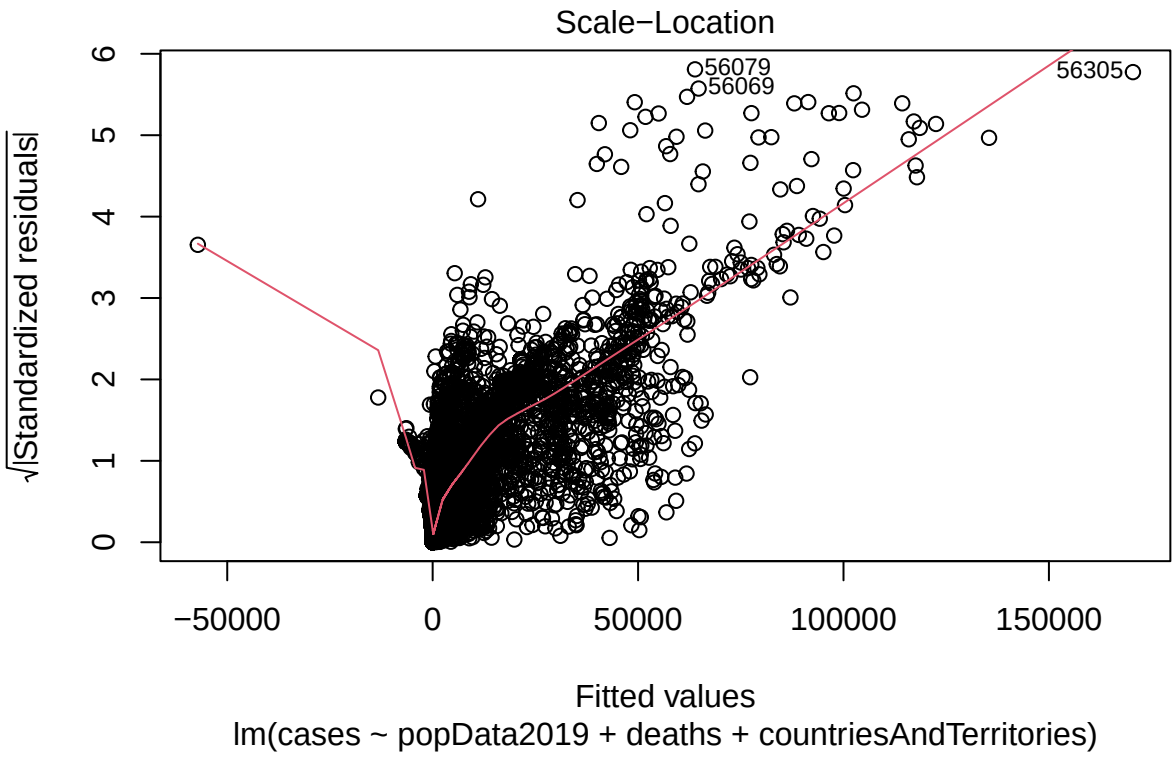
```
## countriesAndTerritoriesMalaysia                                0.775  0.43807
## countriesAndTerritoriesMaldives                                0.025  0.98001
## countriesAndTerritoriesMali                                    0.003  0.99793
## countriesAndTerritoriesMalta                                  -0.018  0.98587
## countriesAndTerritoriesMarshall_Islands                       -0.037  0.97029
## countriesAndTerritoriesMauritania                             -0.019  0.98447
## countriesAndTerritoriesMauritius                              -0.057  0.95417
## countriesAndTerritoriesMexico                                 -4.085 4.42e-05 ***
## countriesAndTerritoriesMoldova                                 0.326  0.74433
## countriesAndTerritoriesMonaco                                 -0.059  0.95294
## countriesAndTerritoriesMongolia                               -0.048  0.96184
## countriesAndTerritoriesMontenegro                              0.116  0.90802
## countriesAndTerritoriesMontserrat                             -0.061  0.95128
## countriesAndTerritoriesMorocco                                 2.261  0.02373 *
## countriesAndTerritoriesMozambique                              0.212  0.83248
## countriesAndTerritoriesMyanmar                                 0.419  0.67554
## countriesAndTerritoriesNepal                                   2.104  0.03541 *
## countriesAndTerritoriesNetherlands                             2.741  0.00612 **
## countriesAndTerritoriesNew_Caledonia                          -0.060  0.95189
## countriesAndTerritoriesNew_Zealand                            -0.047  0.96285
## countriesAndTerritoriesNicaragua                              -0.038  0.96941
## countriesAndTerritoriesNiger                                   0.020  0.98414
## countriesAndTerritoriesNigeria                                 0.158  0.87467
## countriesAndTerritoriesNorth_Macedonia                         0.000  0.99999
## countriesAndTerritoriesNorthern_Mariana_Islands               -0.060  0.95232
## countriesAndTerritoriesNorway                                  0.135  0.89296
## countriesAndTerritoriesOman                                    0.489  0.62455
## countriesAndTerritoriesPakistan                                0.295  0.76769
## countriesAndTerritoriesPalestine                               0.680  0.49662
## countriesAndTerritoriesPanama                                  0.663  0.50754
## countriesAndTerritoriesPapua_New_Guinea                       -0.036  0.97095
## countriesAndTerritoriesParaguay                                0.245  0.80640
## countriesAndTerritoriesPeru                                   -1.360  0.17386
## countriesAndTerritoriesPhilippines                             0.560  0.57543
## countriesAndTerritoriesPoland                                  4.756 1.98e-06 ***
## countriesAndTerritoriesPortugal                                1.564  0.11792
## countriesAndTerritoriesPuerto_Rico                             0.499  0.61759
## countriesAndTerritoriesQatar                                   0.778  0.43660
## countriesAndTerritoriesRomania                                 1.415  0.15713
## countriesAndTerritoriesRussia                                  2.123  0.03376 *
## countriesAndTerritoriesRwanda                                  0.019  0.98453
## countriesAndTerritoriesSaint_Kitts_and_Nevis                  -0.061  0.95167
## countriesAndTerritoriesSaint_Lucia                            -0.060  0.95236
## countriesAndTerritoriesSaint_Vincent_and_the_Grenadines       -0.060  0.95214
## countriesAndTerritoriesSan_Marino                             -0.061  0.95175
## countriesAndTerritoriesSao_Tome_and_Principe                  -0.056  0.95526
## countriesAndTerritoriesSaudi_Arabia                            2.056  0.03982 *
## countriesAndTerritoriesSenegal                                 0.053  0.95752
## countriesAndTerritoriesSerbia                                  1.591  0.11155
## countriesAndTerritoriesSeychelles                             -0.060  0.95241
## countriesAndTerritoriesSierra_Leone                           -0.042  0.96674
## countriesAndTerritoriesSingapore                               0.332  0.73965
## countriesAndTerritoriesSint_Maarten                           -0.057  0.95479
## countriesAndTerritoriesSlovakia                                0.730  0.46555
```

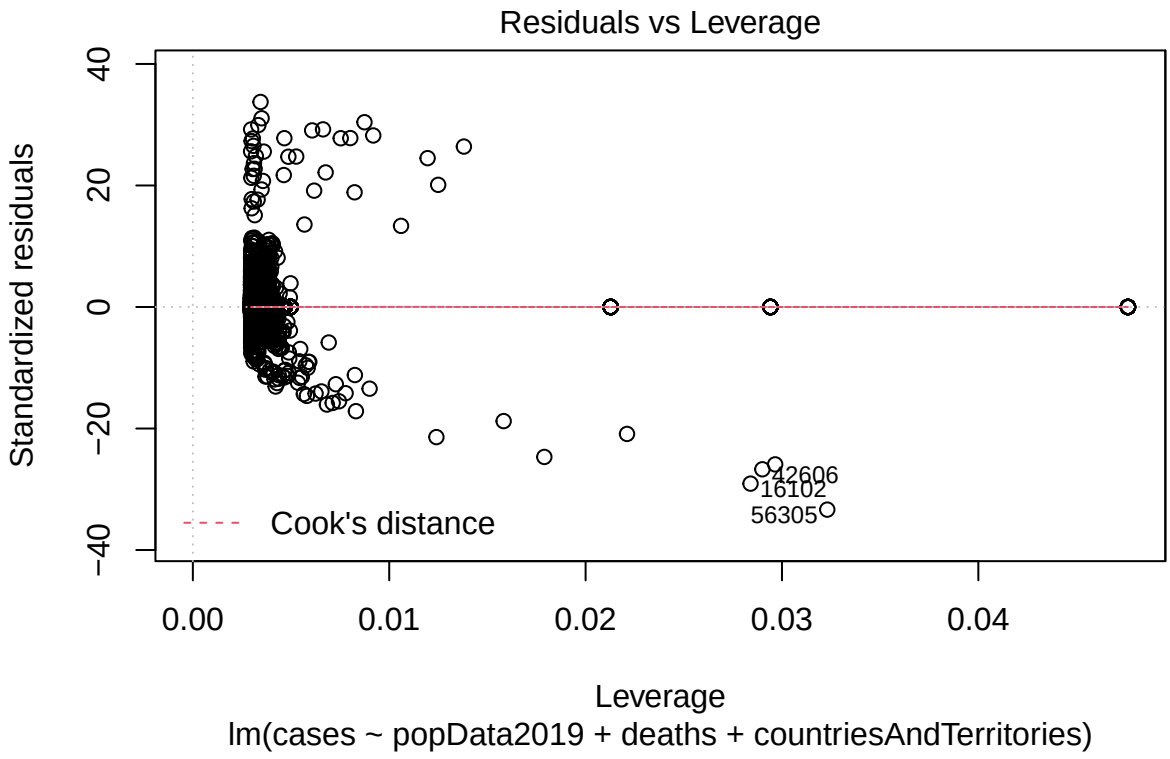
```
## countriesAndTerritoriesSlovenia                                0.324  0.74596
## countriesAndTerritoriesSolomon_Islands                        -0.040  0.96776
## countriesAndTerritoriesSomalia                                -0.008  0.99380
## countriesAndTerritoriesSouth_Africa                            1.147  0.25146
## countriesAndTerritoriesSouth_Korea                             0.282  0.77823
## countriesAndTerritoriesSouth_Sudan                            -0.020  0.98416
## countriesAndTerritoriesSpain                                   2.194  0.02822 *
## countriesAndTerritoriesSri_Lanka                               0.288  0.77371
## countriesAndTerritoriesSudan                                  -0.086  0.93121
## countriesAndTerritoriesSuriname                               -0.047  0.96286
## countriesAndTerritoriesSweden                                  0.656  0.51164
## countriesAndTerritoriesSwitzerland                             1.458  0.14480
## countriesAndTerritoriesSyria                                  -0.081  0.93570
## countriesAndTerritoriesTaiwan                                  0.031  0.97547
## countriesAndTerritoriesTajikistan                              0.067  0.94619
## countriesAndTerritoriesThailand                                0.131  0.89574
## countriesAndTerritoriesTimor_Leste                            -0.058  0.95347
## countriesAndTerritoriesTogo                                   -0.032  0.97425
## countriesAndTerritoriesTrinidad_and_Tobago                    -0.035  0.97169
## countriesAndTerritoriesTunisia                                -0.092  0.92667
## countriesAndTerritoriesTurkey                                  2.285  0.02231 *
## countriesAndTerritoriesTurks_and_Caicos_islands               -0.056  0.95517
## countriesAndTerritoriesUganda                                  0.303  0.76173
## countriesAndTerritoriesUkraine                                 4.252 2.12e-05 ***
## countriesAndTerritoriesUnited_Arab_Emirates                    1.209  0.22683
## countriesAndTerritoriesUnited_Kingdom                         -0.290  0.77162
## countriesAndTerritoriesUnited_Republic_of_Tanzania             0.116  0.90796
## countriesAndTerritoriesUnited_States_of_America                4.755 1.99e-06 ***
## countriesAndTerritoriesUnited_States_Virgin_Islands           -0.052  0.95817
## countriesAndTerritoriesUruguay                                -0.002  0.99829
## countriesAndTerritoriesUzbekistan                              0.730  0.46562
## countriesAndTerritoriesVanuatu                                -0.031  0.97549
## countriesAndTerritoriesVenezuela                               0.991  0.32148
## countriesAndTerritoriesVietnam                                 0.120  0.90418
## countriesAndTerritoriesWestern_Sahara                         -0.052  0.95813
## countriesAndTerritoriesYemen                                  -0.158  0.87465
## countriesAndTerritoriesZambia                                  0.072  0.94289
## countriesAndTerritoriesZimbabwe                                   NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4276 on 58547 degrees of freedom
## Multiple R-squared:  0.623,  Adjusted R-squared:  0.6217
## F-statistic: 458.6 on 211 and 58547 DF,  p-value: < 2.2e-16
```

```
plot(model)
```


Residuals vs Fitted
Residuals
150000
50000
−50000
−150000
56079
56069
56305
−50000
0
50000
100000
150000
Fitted values
lm(cases ~ popData2019 + deaths + countriesAndTerritories)


Normal Q−Q
Standardized residuals
40
20
0
−20
56079
56069
56305
−4
−2
0
2
4
Theoretical Quantiles
lm(cases ~ popData2019 + deaths + countriesAndTerritories)


Scale−Location
√|Standardized residuals|
0
1
2
3
4
5
6
56079
56069
56305
−50000
0
50000
100000
150000
Fitted values
lm(cases ~ popData2019 + deaths + countriesAndTerritories)



The correlation coefficient is 0.623. This indicates a moderate positive linear relationship between the variables.The regression predictions move in the same direction and fit the data and indicate a strong correlation at a global level.

```
model_cases <- auto.arima(covid_19$cases)
forecast(model_cases,5)
```

```
##       Point Forecast     Lo 80    Hi 80     Lo 95    Hi 95
## 58760       19.76099 -1932.985 1972.507 -2966.706 3006.228
## 58761       33.62798 -2377.251 2444.507 -3653.494 3720.749
## 58762       46.62973 -2691.775 2785.035 -4141.399 4234.658
## 58763       60.39574 -2971.855 3092.646 -4577.031 4697.822
## 58764       72.90229 -3201.969 3347.773 -4935.581 5081.385
```

```
model_deaths <- auto.arima(covid_19$deaths)
forecast(model_deaths,5)
```

```
##       Point Forecast     Lo 80     Hi 80     Lo 95    Hi 95
## 58760       1.523009 -79.35709  82.40311 -122.1724 125.2184
## 58761       2.224514 -86.82285  91.27188 -133.9617 138.4107
## 58762       2.795233 -91.26872  96.85918 -141.0632 146.6536
## 58763       3.270434 -94.11986 100.66073 -145.6752 152.2160
## 58764       3.767232 -97.13330 104.66776 -150.5468 158.0813
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.